

Network Modeling: Assignment 1

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Group 30:

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Task 1

- (1) The total number of possible directed ties is

$$n(n-1) = 462.$$

Therefore, the probability of observing a directed tie is

$$p = \frac{134}{462} = 0.2900.$$

- (2) As there are 231 dyads, the probabilities are

$$p_M = \frac{42}{231} = 0.1818, \quad p_A = \frac{50}{231} = 0.2165, \quad p_N = \frac{139}{231} = 0.6017.$$

- (3) With tie independence, each directed tie is present with probability p . so,

$$p_M^{\text{ind}} = p^2 = 0.0841,$$

$$p_A^{\text{ind}} = 2p(1-p) = 0.4118,$$

$$p_N^{\text{ind}} = (1-p)^2 = 0.5040.$$

- (4) Assuming independence of ties does not hold up in this case, since the calculated probabilities differ significantly from the observed ones. Mutual and null dyads occur more often than expected, while asymmetric dyads occur less often. Reciprocal ties seem to be much more common than would be expected under an independent process.

- (5) If each pupil is assumed to have exactly 9 outgoing ties, then the probability of a directed tie is

$$p = \frac{9}{21} = 0.4286.$$

This gives

$$p_M^{\text{ind}} = p^2 = \left(\frac{9}{21}\right)^2 = 0.1837.$$

Compared to the random chance independent p_M^{ind} we get a very close estimate of p_M with our assumption of 9 outgoing edges. The model is very reasonable since pupils might be friends with on average 9 other pupils, although this would probably not be uniformly distributed.

- (6) For **cguRec**, the hypothesis is that the observed reciprocity is higher than expected by chance, with the number of edges fixed. Since $\Pr(X \geq \text{Obs}) = 0$, none of the random graphs had a reciprocity as high as the observed one 0.7737. This strongly suggests that friendships are much more often mutual than one-sided.

For **cugInd**, the hypothesis is that the observed indegree centralization is higher than expected by chance, again with the number of edges fixed. Since $\Pr(X \geq \text{Obs}) = 0.7712$, the observed value (0.2022) is not unusually high. This suggests that there is no single pupil or small group of pupils that is much more popular than expected by chance.

For **cugTrans**, the hypothesis is that the observed transitivity is higher than expected by chance, with the dyad census fixed. Since $\Pr(X \geq \text{Obs}) = 0$, none of the random graphs had a transitivity as high as the observed one (0.5360). This strongly suggests that pupils tend to be friends with the friends of their friends.

Overall, the network shows clearly higher reciprocity and transitivity than expected by chance, while indegree centralization is not unusually high.

Task 2

(2.1) The QAP gives a coefficient of 0.9998. This implies a very strong positive association between the friendship network at wave 1 and the friendship network at wave 2. The corresponding p-value is 0.0, showing, that the result is also very significant.

(2.2) We added three dyadic covariate matrices based on the attribute data.

First, we used the literacy score of the receiver. For this, the literacy value of each student was copied into the corresponding column of the matrix, so that the receiver's literacy score can affect the probability of receiving a friendship tie.

Second, we created a same-gender matrix. Its entries are 1 if two students have the same gender and 0 otherwise.

Third, we used the HISEI value of the sender. For this, the HISEI value of each student was copied into the corresponding row of the matrix, so that the sender's HISEI can affect the probability of sending a friendship tie.

(2.3) **Hypothesis i:**

The coefficient is negative, but very small and not statistically significant. So there is no meaningful support for the hypothesis.

Hypothesis ii:

The coefficient is negative and statistically significant. So this hypothesis is supported.

Hypothesis iii:

The coefficient is negative and not statistically significant. So there is no support for the hypothesis.

(2.4) See code.

(2.5) See table for the estimated values.

Hypothesis i: The coefficients are all very close to zero and can be either positive or negative. This shows no support for the hypothesis that students with high literacy scores receive fewer nominations.

Hypothesis ii: The coefficients are mostly small, with both positive and negative values. This suggests that same-gender friendships may be more likely in some classrooms, but the effect is not consistent across all classrooms.

Hypothesis iii: The coefficients are again very small and can be positive or negative. This shows no evidence that students with higher HISEI scores send more friendship nominations.

	classroom	intercept	w1	literacy	gender	HISEI
1	01	0.0269	0.9999	-0.0013	0.0209	0.0009
2	02	-0.1872	0.9999	-0.0015	0.0440	0.0034
3	03	-0.0890	1.0001	0.0019	0.0886	-0.0010
4	04	-0.1279	0.9999	0.0016	-0.0557	0.0022
5	05	0.2420	0.5777	0.0009	-0.0421	-0.0017
6	10	0.5139	0.9997	-0.0012	-0.1339	-0.0030
7	12	-0.1734	0.9999	0.0042	0.0462	0.0023

Figure 1: QAP coefficients across classrooms.

Task 3

(3.1) See code.

(3.2) See code.

(3.3) In the NAM model for class 10 wave 2, the network autocorrelation parameter (ρ_1) was estimated at 0.613 and is highly statistically significant ($p < 0.001$). Thus, the data strongly supports the hypothesis that a student's ending literacy score is positively associated with the literacy scores of their friends.

Furthermore, when controlling for these network effects, the exogenous variables gender ($p = 0.199$) and socioeconomic status/HISEI ($p = 0.902$) were not statistically significant predictors of a student's ending literacy score.

(3.4) With the comparison between MR-QAP and NAM, we can observe the difference between social ties and individual outcomes. In MR-QAP, gender homophily was highly significant predictor for friendship ties ($p < 0.001$). Thus, students tended to form same-gender groups. However, in NAM, gender was not statistically significant for predicting the student's ending literacy score.

When it comes to socioeconomic status, both NAM and MR-QAP didn't display statistical significance: HISEI did not affect a student's consideration for friendship ($p = 0.658$), nor did it predict high ending literacy score in NAM ($p = 0.902$).

(3.6) Analyzing the network autocorrelation parameter, we see that classrooms 2, 3, 4, 8, 10 and 11 have positive estimates, implying that overall, friendships are correlated with similar literacy scores. However, this cannot be generalized since only classrooms 3, 10 and 11 have strong statistical significance. For the rest of the classrooms, the network effect is not strong enough to be discerned from the random noise. While there is a positive trend, the statistical significance of the autocorrelation hypothesis is not generalizable across all classrooms.

Code

task1.R

```

1 # task 1:
2 #Group 30: Tristan Gabl, Hristo Georgiev, Kevin Halter, Jakob Teetz
3
4 seed = 161
5 set.seed(seed)
6 num_perm = 5000
7
8 # (1)
9 # 22 pupils -> total number of possible dyads =  $n*(n-1)/2 = 22*21/2 = 231$ 
10 # for (3) to make sense though, we will use "probability p of observing a tie if
    you choose a directed edge at random"
11 # number of ties m = 134, number of directed edges =  $2 * 231 = 462$ 
12 num_directed = 462
13 p = 134 / num_directed
14 print(paste("task1.1: p =", p))
15
16 # (2)
17 # M = 42, A = 50, N = 139. sum = 231
18 p_M = 42 / 231
19 p_A = 50 / 231
20 p_N = 139 / 231
21 print(paste("task1.2: p_M =", p_M))
22 print(paste("task1.2: p_A =", p_A))
23 print(paste("task1.2: p_N =", p_N))
24
25 # (3)
26 # it means we assume for a mutual tie for example that  $p(\text{mutual}) = p(i \rightarrow j) * p(j \rightarrow i)$ 
    =  $p^2$ 
27 p_M_ind = p*p
28 p_A_ind = 2*p*(1-p)
29 p_N_ind = (p-1)*(p-1)
30 print(paste("task1.3: p_M_ind =  $p^2$  =", p_M_ind))
31 print(paste("task1.3: p_A_ind =  $2*p*(1-p)$  =", p_A_ind))
32 print(paste("task1.3: p_N_ind =  $(1-p)^2$  =", p_N_ind))
33
34 # (4)
35 print("task1.4:")
36 print("The results show:  $p\_M\_ind < p\_M$ ,  $p\_A < p\_A\_ind$ ,  $p\_N\_ind < p\_N$ .")
37 print("From that we can conclude that the ties are not independent, since mutual
    connection as well as no connection is much more probable compared to the
    independent case, assymmetric dyads are half as probable compared")
38
39 # (5)
40 # we have 22 pupils, if each pupil now is assumed to make 9 outward connections we
    get  $p = 9 / 21$ 
41 p_new = 9/21
42 p_M_ind_new = p_new*p_new
43 print(paste("task1.5: p_new =", p_new))
44 print(paste("task1.5: p_M_ind_new =", p_M_ind_new))

```

```

45 print("task1.5: compared to the random chance independent p_M we get a very close
    estimate of p_M with our assumption of 9 outgoing edges. The model is very
    reasonable since pupils might be friends with on average 9 other pupils.")
46
47
48 # (6)
49 install.packages("sna")
50 install.packages("network")
51 library(sna)
52 library(network)
53
54 obsMat = read.csv2("./Lintner/10_W1.csv", header = TRUE, row.names = 1)
55 print(dim(obsMat))
56 print(obsMat)
57 obsMat_clean = obsMat[!is.na(rowSums(obsMat)),!is.na(rowSums(obsMat))]
58 print(dim(obsMat_clean))
59 print(obsMat_clean)
60 obsMat_clean = as.matrix(obsMat_clean)
61 print(obsMat_clean)
62
63 # cug.test(dat, FUN, mode = c("digraph", "graph"), cmode = c("size",
64 #     "edges", "dyad.census"), diag = FALSE, reps = 1000,
65 #     ignore.eval = TRUE, FUN.args = list())
66
67 # FUN = function generating the test statistic (grecip -> the dyadic reciprocity
68 #     of a graph is the proportion of dyads which are symmetric;
69 #     centralization -> returns
70 #     centralization GLI given a node centrality measure
71 #     gtrans -> computes the
72 #     transitivity of a graph)
73
74 # cmode = string indicating the type of conditioning to be applied (edges -> holds
75 #     no. of edges fixes; dyad.census -> holds no. of dyad fixed as observed)
76
77
78 cguRec <- cug.test(obsMat_clean, grecip, cmode = "edges", reps=num_perm)
79 cugInd <- cug.test(obsMat_clean, centralization, cmode="edges", FUN.arg=list(FUN=
80     degree, cmode="indegree"), reps=num_perm)
81 cugTrans <- cug.test(obsMat_clean, gtrans, cmode = "dyad.census", reps=num_perm)
82
83 # (6.1)
84 #
85 print("cguRec: hypotheses -> The observed reciprocity is higher than expected
86     compared to chance; conditional feature -> number of edges is fixed")
87 print("cugInd: hypotheses -> The observed indegree centralization is higher than
88     expected compared to chance; conditional feature -> number of edges is fixed")
89 print("cugTrans: hypotheses -> The observed transitivity is higher than expected
90     compared to chance; conditional feature -> Dyad census (observed M/A/N counts)
91     is fixed")
92
93 # (6.2)
94 print(cguRec)
95 print(cugInd)

```

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87 print(cugTrans)
88
89 print("task1.6.2: cguRec -> Pr(X>=Obs) = 0, Pr(X<=Obs) = 1, non of the random
    graphs had a higher observed reciprocity (0.77), therefore the data strongly
    suggests pupils are friends mutually instead of just one sided")
90 print("task1.6.2: cugInd -> Pr(X>=Obs) = 0.7746, Pr(X<=Obs) = 0.6152, data does
    not show a pupil being overly popular, popularity is spread evenly")
91 print("task1.6.2: cugTrans -> Pr(X>=Obs) = 0, Pr(X<=Obs) = 1, non of the random
    graphs had a higher observed transistivity (0.536), therefore the data strongly
    suggests puils are friends with the friends of friends")

```

task2.R

```

1  #task 2
2  #Group 30: Tristan Gabl, Hristo Georgiev, Kevin Halter, Jakob Teetz
3
4  library(readr)
5  library(sna)
6  library(tidyverse)
7  library(dplyr)
8  library(gridExtra)
9  library(grid)
10
11 #setwd("./Lintner")
12
13 #Task 2.1
14 attr <- read.csv("attr.csv", sep = ";")
15
16 w1 <- as.matrix(read.csv("10_W1.csv", header = FALSE, sep = ";"))
17 w2 <- as.matrix(read.csv("10_W2.csv", header = FALSE, sep = ";"))
18
19 attr10 <- attr %>% filter(classroomID == 10)
20
21 complete_idx <- which(complete.cases(attr10))
22 attr10 <- attr10[complete_idx, ]
23 w1 <- w1[complete_idx, complete_idx]
24 w2 <- w2[complete_idx, complete_idx]
25
26 model1 <- netlm(w2, w1, nullhyp = "qap", reps = 5000)
27
28 summary(model1)
29
30
31 #Task 2.2 / 2.3:
32
33 literacy_matrix <- outer(rep(1, nrow(attr10)), attr10$literacy_end)
34 gender_matrix <- outer(attr10$gender, attr10$gender, FUN = "==") * 1
35 HISEI_matrix <- outer(attr10$HISEI, rep(1, nrow(attr10)))
36
37 model2 <- netlm(
38   w2,
39   list(w1, literacy_matrix, gender_matrix, HISEI_matrix),

```

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40   nullhyp = "qap",
41   reps = 5000
42 )
43
44 summary(model2)
45
46
47 #Task 2.4 / 2.5
48
49 classrooms <- c("01", "02", "03", "04", "05", "10", "12")
50 results <- list()
51
52 for (class in classrooms) {
53   w1 <- as.matrix(read.csv(paste0(class, "_W1.csv"), header = FALSE, sep = ";"))
54   w2 <- as.matrix(read.csv(paste0(class, "_W2.csv"), header = FALSE, sep = ";"))
55   attr <- read.csv("attr.csv", sep = ";")
56
57   attr_class <- attr %>% filter(classroomID == as.numeric(class))
58
59   complete_idx <- which(complete.cases(attr_class))
60   attr_class <- attr_class[complete_idx, ]
61   w1 <- w1[complete_idx, complete_idx]
62   w2 <- w2[complete_idx, complete_idx]
63
64   literacy_matrix <- outer(rep(1, nrow(attr_class)), attr_class$literacy_end)
65   gender_matrix <- outer(attr_class$gender, attr_class$gender, FUN = "==") * 1
66   HISEI_matrix <- outer(attr_class$HISEI, rep(1, nrow(attr_class)))
67
68   model <- netlm(
69     w2,
70     list(w1, literacy_matrix, gender_matrix, HISEI_matrix),
71     nullhyp = "qap",
72     reps = 5000
73   )
74
75   results[[class]] <- summary(model)$coefficients
76 }
77
78 results
79
80
81 #Task 2.5:
82
83 summary_df <- bind_rows(
84   lapply(names(results), function(class) {
85     data.frame(
86       classroom = class,
87       intercept = results[[class]][1],
88       w1 = results[[class]][2],
89       literacy = results[[class]][3],
90       gender = results[[class]][4],
91       HISEI = results[[class]][5]
92     )

```



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93   })
94   )
95
96   summary_df
97
98   summary_df_round <- summary_df
99   summary_df_round[-1] <- lapply(summary_df_round[-1], function(x) round(x, 4))
100
101   png("../task2_table.png", width = 1200, height = 400, res = 150)
102   grid.table(summary_df_round)
103   dev.off()

```

task3.R

```

1  #task 3
2  #Group 30: Tristan Gabl, Hristo Georgiev, Kevin Halter, Jakob Teetz
3
4  library(sna)
5  library(network)
6  library(ggplot2)
7
8  seed = 161
9  set.seed(seed)
10
11 # load 10_W2
12 net_w2 <- read.csv("../Lintner/10_W2.csv", header = TRUE, row.names = 1, sep = ";")
13 net_w2_mat <- as.matrix(net_w2)
14
15 # load attr for classroom 10
16 attributes <- read.csv("../Lintner/attr.csv", header = TRUE, sep = ";")
17 attr_10 <- attributes[attributes$classroomID == 10, ]
18
19 node_ids <- gsub("[^0-9]", "", rownames(net_w2_mat))
20 attr_ids <- gsub("[^0-9]", "", as.character(attr_10$studentID))
21
22 # match the cleaned IDs
23 attr_10 <- attr_10[match(node_ids, attr_ids), ]
24
25 # convert gender to a binary for regression
26 attr_10$gender_female <- ifelse(attr_10$gender == "female", 1, 0)
27 cat("1. Starting Network dimensions:", nrow(net_w2_mat), "rows by", ncol(net_w2_
    mat), "columns\n")
28 valid_nodes_attr <- complete.cases(attr_10[, c("literacy_end", "gender_female", "
    HISEI")])
29 cat("2. Nodes with valid attributes:", sum(valid_nodes_attr, na.rm = TRUE), "\n")
30 diag(net_w2_mat) <- 0
31 valid_nodes_net <- rowSums(is.na(net_w2_mat)) < (ncol(net_w2_mat) - 1)
32 cat("3. Nodes with valid network data:", sum(valid_nodes_net, na.rm = TRUE), "\n")
33
34 valid_nodes <- valid_nodes_attr & valid_nodes_net
35
36 net_w2_clean <- net_w2_mat[valid_nodes, valid_nodes]

```

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37 attr_10_clean <- attr_10[valid_nodes, ]
38
39 cat("\nStudents remaining after cleaning:", nrow(net_w2_clean), "\n\n")
40
41 # 3.1
42 y <- attr_10_clean$literacy_end
43 W_raw <- net_w2_clean
44
45 # row normalization trick
46 row_sums <- rowSums(W_raw)
47 row_sums[row_sums == 0] <- 1
48 W_norm <- W_raw / row_sums
49
50 nam_model_1 <- lnam(y, W1 = W_norm)
51 print("task 3.1")
52 summary(nam_model_1)
53
54 # 3.2
55 X <- cbind(attr_10_clean$gender_female, attr_10_clean$HISEI)
56 colnames(X) <- c("Gender_Female", "HISEI")
57
58 # 3.3
59 # interpretation: See task3.tex.
60 nam_model_2 <- lnam(y, x = X, W1 = W_norm)
61 print("task 3.3")
62 summary(nam_model_2)
63
64 # 3.4 is dependent on task 2. See task3.tex
65
66 # 3.5
67 run_nam_for_classroom <- function(class_id, attr_data) {
68   file_name <- sprintf("./Lintner/%02d_W2.csv", class_id)
69
70   if(!file.exists(file_name)) {
71     return(paste("File", file_name, "not found."))
72   }
73
74   net_mat <- as.matrix(read.csv(file_name, header = TRUE, row.names = 1, sep = ";"))
75   attr_sub <- attr_data[attr_data$classroomID == class_id, ]
76
77   node_ids <- gsub("[^0-9]", "", rownames(net_mat))
78   attr_ids <- gsub("[^0-9]", "", as.character(attr_sub$studentID))
79   attr_sub <- attr_sub[match(node_ids, attr_ids), ]
80
81   attr_sub$gender_female <- ifelse(attr_sub$gender == "female", 1, 0)
82
83   v_attr <- complete.cases(attr_sub[, c("literacy_end", "gender_female", "HISEI")])
84   diag(net_mat) <- 0
85   v_net <- rowSums(is.na(net_mat)) < (ncol(net_mat) - 1)
86
87   valid <- v_attr & v_net

```

```

88
89 net_clean <- net_mat[valid, valid]
90 attr_clean <- attr_sub[valid, ]
91
92 # model
93 y <- attr_clean$literacy_end
94 W_raw <- net_clean
95
96 #row normalization trick
97 row_sums <- rowSums(W_raw)
98 row_sums[row_sums == 0] <- 1
99 W_norm <- W_raw / row_sums
100
101 X <- cbind(attr_clean$gender_female, attr_clean$HISEI)
102 colnames(X) <- c("Gender_Female", "HISEI")
103
104 model <- lnam(y, x = X, W1 = W_norm)
105 return(summary(model))
106 }
107
108 # run on all unique classrooms
109 classrooms <- unique(attributes$classroomID)
110
111 # get rid of classroom 10 since we already analysed it
112 remaining_classrooms <- classrooms[!(classrooms %in% c(10,6,7,8,11))]
113
114 for(cid in remaining_classrooms) {
115   cat("\n===== \n")
116   cat(" Task 3.5: NAM Results for Classroom:", cid, "\n")
117   cat("===== \n")
118
119   result <- run_nam_for_classroom(cid, attributes)
120   print(result)
121 }
122
123 #3.6
124 # analysis: See task3.tex
125 plot_classrooms <- classrooms[!(classrooms %in% c(10,6,7,8,11))]
126
127 class_ids <- c()
128 rho_ests <- c()
129 rho_ses <- c()
130
131 for (cid in plot_classrooms) {
132   file_name <- sprintf("./Lintner/%02d_W2.csv", cid)
133
134   if (file.exists(file_name)) {
135     net_mat <- as.matrix(read.csv(file_name, header = TRUE, row.names = 1, sep = "
;"))
136     attr_sub <- attributes[attributes$classroomID == cid, ]
137
138     node_ids <- gsub("[^0-9]", "", rownames(net_mat))
139     attr_ids <- gsub("[^0-9]", "", as.character(attr_sub$studentID))

```

```

140   attr_sub <- attr_sub[match(node_ids, attr_ids), ]
141   attr_sub$gender_female <- ifelse(attr_sub$gender == "female", 1, 0)
142
143   v_attr <- complete.cases(attr_sub[, c("literacy_end", "gender_female", "HISEI"
144     )])
145   diag(net_mat) <- 0
146   v_net <- rowSums(is.na(net_mat)) < (ncol(net_mat) - 1)
147
148   valid <- v_attr & v_net
149   net_clean <- net_mat[valid, valid]
150   attr_clean <- attr_sub[valid, ]
151
152   if (nrow(net_clean) > 0) {
153     y <- attr_clean$literacy_end
154     W_raw <- net_clean
155
156     row_sums <- rowSums(W_raw)
157     row_sums[row_sums == 0] <- 1
158     W_norm <- W_raw / row_sums
159
160     X <- cbind(attr_clean$gender_female, attr_clean$HISEI)
161     colnames(X) <- c("Gender_Female", "HISEI")
162
163     model <- lnam(y, x = X, W1 = W_norm)
164
165     class_ids <- c(class_ids, cid)
166     rho_ests <- c(rho_ests, model$rho1)
167     rho_ses <- c(rho_ses, model$rho1.se)
168   }
169 }
170
171 plot_data <- data.frame(
172   Classroom = as.factor(class_ids),
173   Rho = rho_ests,
174   SE = rho_ses
175 )
176
177 # 0.95 conf interval
178 plot_data$Lower <- plot_data$Rho - 1.96 * plot_data$SE
179 plot_data$Upper <- plot_data$Rho + 1.96 * plot_data$SE
180
181 # plot
182 ggplot(plot_data, aes(x = Classroom, y = Rho)) +
183   geom_point(size = 3, color = "blue") +
184   geom_errorbar(aes(ymin = Lower, ymax = Upper), width = 0.2, color = "darkblue")
185   +
186   geom_hline(yintercept = 0, linetype = "dashed", color = "red", size = 1) +
187   coord_flip() +
188   labs(title = "Network Autocorrelation (\u03c1) Across Classrooms",
189     subtitle = "With 95% Confidence Intervals",
190     x = "Classroom ID",
191     y = "Autocorrelation Estimate (\u03c1)") +

```

```
191 | theme_minimal()
```